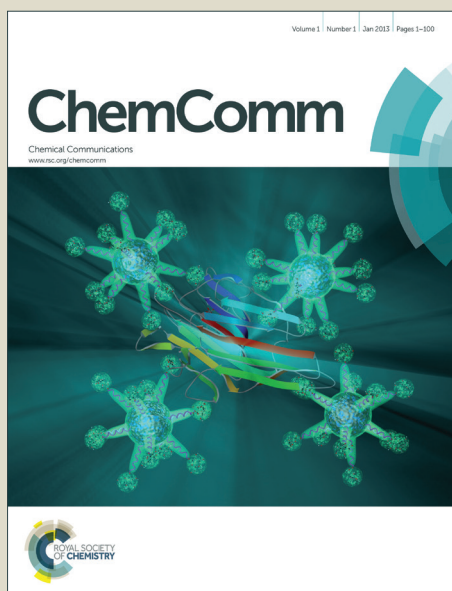


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ARTICLE TYPE

Synthesis of Diarylsulfones with Simple Arenes and $K_2S_2O_8$ through Double C-S Bond Formation

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An unprecedented double C-S bond formation method has been developed to prepare both symmetric and unsymmetric diarylsulfones with simple arenes in a single step. This represents the first example that $K_2S_2O_8$ can be employed as a highly effective sulfonating agent to synthesize diarylsulfones. The reaction demonstrates excellent reactivity, good functional group tolerance and high yields.

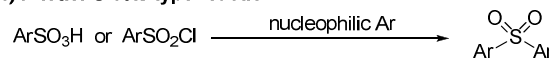
Diarylsulfone derivatives are an important class of chemicals which find broad utility in pharmaceutical and polymer industries.¹ Compounds containing a diarylsulfone scaffold are being developed in a wide range of therapeutic areas including leprosy, depression and cardiac diseases.² To date, a number of diarylsulfone-containing molecules has been successfully commercialized, such as Dapsone, SR 33805 oxalate and CX157.³ Due to their important applications, the synthesis of diarylsulfone has been extensively investigated (scheme 1). By far the most common strategies towards diarylsulfone synthesis typically involve the classic Friedel-Crafts type sulfonation reactions⁴ and oxidation of diarylsulfides⁵. These protocols generally suffer from significant limitations, such as harsh conditions, lack of regioselectivity or the use of thiols with unpleasant odors. In the past decade, the cross-coupling of arenesulfonate salts with arenehalides or areneboronic acids through transition-metal catalysis⁶ or under metal-free conditions^{3b, 7} provides a convenient and efficient route to access diarylsulfones. However, those coupling methods usually have one or more innate drawbacks which include narrow substrate scope, high temperature, limited availability of sodium sulfonates^{3a, 8}, the use of stoichiometric additives, and tedious reaction procedures, etc. Although a number of new methods have been developed in recent years⁹, developing a general, efficient and complementary synthetic protocol to diarylsulfone is still in great demand. Herein, we report the development of a facile and practical method with $K_2S_2O_8$ as an effective sulfonating agent for preparation of symmetric diarylsulfones in one step and unsymmetrical diarylsulfones in one pot reaction.

In our continuous studies of developing new tools to access highly functionalized arenes, we are particularly interested in the development of an alternative transition-metal-free approach towards diarylsulfone synthesis. Direct transformation of readily available arenes into valuable complex diarylsulfones with a

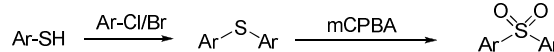
specific sulfur-containing oxidant is arguably a highly efficient method toward these compounds. Our goal was to identify an ideal sulfur-containing oxidant which should have a few merits such as good reactivity, relative low price, easy and safe to handle, a long shelf life, etc.

Scheme 1. General approaches to access diarylsulfones.

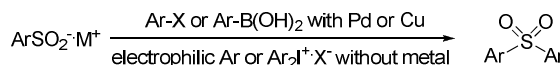
1) Friedel-Crafts type reaction



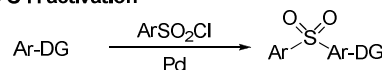
2) Oxidation of sulfide



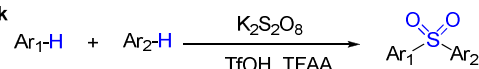
3) Coupling of sulfinate salt



4) Ortho C-H activation



This work



At the beginning of our work, persulfate salts attract our attentions because historically these chemicals had been broadly used as strong oxidants in bleaching¹⁰, polymerization¹¹ and many different organic reactions¹². In addition, persulfate salts are generally very stable and cheap chemical reagents. However, to the best of our knowledge, the employment of persulfate salts as sulfonating agents remains unexplored. We envisioned that persulfate salts like potassium persulfate, under proper acidic conditions, might promote a sequential oxidative C-S bond formation with two molecules of arene to provide corresponding diarylsulfone products. Our aim was to identify suitable conditions with persulfate salts as practical sulfur sources to access diarylsulfones with simple arenes.

Herein, as briefly shown in table 1, we evaluated numerous conditions with varied combinations of persulfate salts and acids by employing unsubstituted benzene **1** as a model compound. After many fruitless attempts, to our delight, the desired product **2** was observed in 39% yield after stirring for 6 hrs at 80°C in the presence of potassium persulfate, triflic acid (TfOH) and trifluoroacetic anhydride (TFAA) (entry 1). Further investigations

revealed that both TFOH and TFAA were essential for this reaction. The absence of either one of the reagents will result the failure of the reactions (entry 2, 3, 4 and 7). The proper amounts of TFOH (4.0 equiv.) and TFAA (11.0 equiv.) were found to be critical for the reaction as well (entry 5). Other persulfate salts, such as $\text{Na}_2\text{S}_2\text{O}_8$ and $(\text{NH}_4)_2\text{S}_2\text{O}_8$, also can promote the reactions and give similar results (entry 9, 10). Considering the heterogeneous and sticky appearance of the reaction system, tetrabutylammonium hydrogen sulfate used as phase transfer catalyst (PTC) were added¹³. Pleasingly, further optimizations can improve the yield up to 81% (entry 12). *It is noteworthy to point out that the yields are calculated based on the fact that one $\text{K}_2\text{S}_2\text{O}_8$ molecule will provide all the two sulfur atoms* (for example, if 0.2mmol $\text{K}_2\text{S}_2\text{O}_8$ are used, 0.4mmol diarylsulfone products will be expected), which indicates that $\text{K}_2\text{S}_2\text{O}_8$ is a highly atom-efficient sulfonating agent which has not been discovered before¹⁴.

Table 1. Optimization of reaction conditions^a.

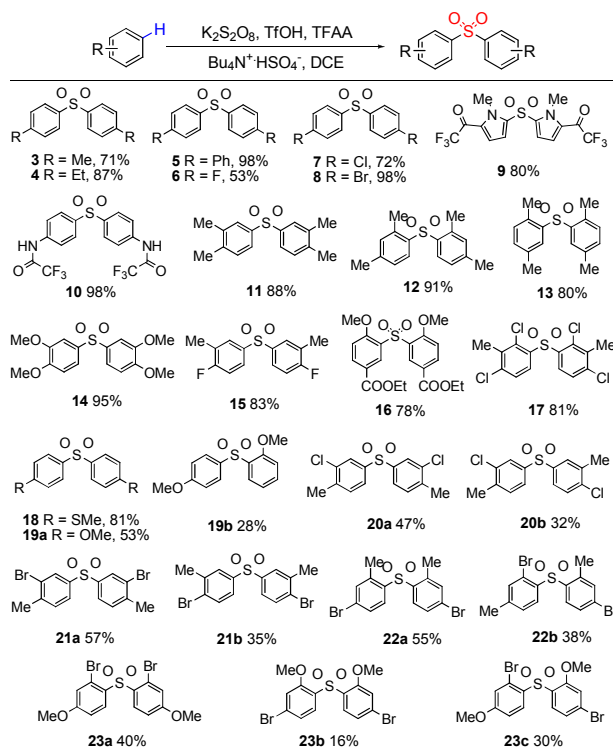
entry	conditions ^a	yield ^b (%)
1	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 2.0eq, TFAA 11.0eq	39%
2	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, AcOH 4.0eq, TFAA 11.0eq	NR ^d
3	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFA 4.0eq, TFAA 11.0eq	NR
4	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, MeSO_3H 4.0eq, TFAA 11.0eq	NR
5	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 11.0eq, 3h	47%
6	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 11.0eq, 60°C	50%
7	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, Ac_2O 11.0eq	2%
8	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 11.0eq	57% ^c
9	$\text{Na}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 11.0eq	51%
10	$(\text{NH}_4)_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 11.0eq	59%
11	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 4.0eq	1%
12	$\text{K}_2\text{S}_2\text{O}_8$ 1.0eq, TFOH 4.0eq, TFAA 11.0eq, PTC 0.1eq	81% ^c

^aThe reaction was run at 0.2mmol scale, 5.0eq benzene, 0.3ml 1,2-dichloroethane (DCE) at 80°C for 6h. PTC : $(\text{Bu}_4\text{N}^+\text{HSO}_4^-)$. ^bNMR yield. ^cIsolated yield ^dNo reaction. More details see SI.

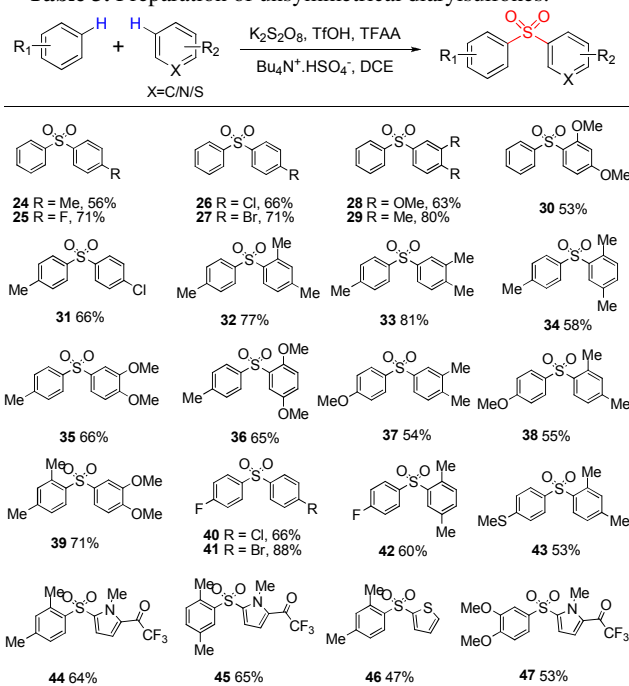
With the optimal conditions in hand, we next investigated the reaction scope for the synthesis of symmetric diarylsulfone. As displayed in table 2, a variety of substituted arenes were efficiently transformed into the corresponding symmetric diarylsulfone products (**3-23**) in modest to excellent yields by tuning the temperature, the amounts of acid and anhydride specifically. The scope of the substituents was found to be quite broad. Both mono-substituted and multi-substituted aryl groups, as well as the electron donating and electron withdrawing functional groups (methyl, methoxyl, ester and halides, etc.) were well tolerated. It was found that regioselectivity outcomes were mainly controlled by the electronic effect. In most cases of using mono-substituted arenes (**3-18**), one para regioisomer were obtained as the major product and only trace amounts of other regioisomers can be observed. As to multi-substituted arenes (**20-23**), more than one regioisomers can be oftenly obtained, which resulted from the balance of electronic character of substituent groups. It is noteworthy to emphasize that some products substituted with halides, such as compounds **6-8**, **15** and **17**, can be further modified to provide a broad variety of valuable derivatives through transition metal catalyzed C-C, C-N, C-O cross-coupling reactions¹⁵. Pleasingly, heterocycle N-methyl pyrrole can be employed in this reaction to provide corresponding trifluoroacetylated sulfone **9** as well. In addition, it is notable that

the employment of 2, 2, 2-trifluoro-N-phenyl acetamide as substrate will provide sulfone product **10** which can undergo one step further transformation to provide the drug dapsone¹⁶, an antibacterial agent. Interestingly, methyl(phenyl)sulfane can be converted into expected sulfone **18** without the potential oxidation of the sulfur ether.

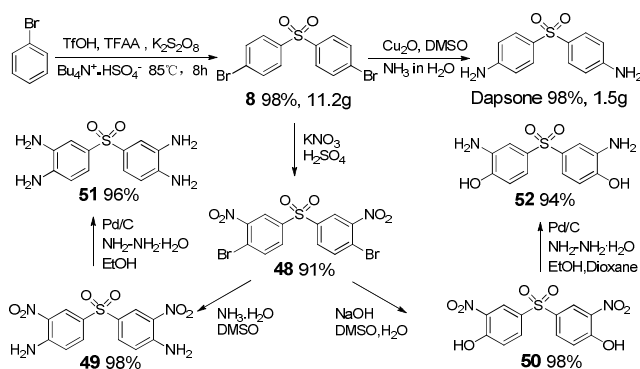
Table 2. Preparation of symmetric diarylsulfones.



The success of symmetric diarylsulfone synthesis encouraged us to further examine the potential utility of this new method in preparation of highly valuable and more challenging unsymmetric diarylsulfones. Direct applying previous optimum conditions proved unsuccessful. It was found that the premixed two different arenes would always give the symmetric diarylsulfone of the more electron-rich arene, but not desired unsymmetric diarylsulfone, as the major products. To solve this problem, further conditions optimization were extensively attempted. Finally, a new one-pot protocol was established. Instead of a premixture of two arene substrates in previous operation, the major change of the new procedure is that one arene substrate, $\text{K}_2\text{S}_2\text{O}_8$, TFOH, PTC and solvent will be charged to the reaction firstly, then, another arene and TFAA would be added after a certain period of time (for details please see SI). As demonstrated in table 3, to our delight, a variety of unsymmetric diarylsulfones can be efficiently synthesized in good to excellent yields (**24-47**). Diarylsulfone products containing two different halide atoms, like **40** and **41**, can provide convenience for further chemical transformations. Remarkably, not only substituted arenes, but also heteroarenes including pyrrole and thiophene can be used to furnish desired products (**44-47**). It is important to point out all these unsymmetric diarylsulfones would be either difficult or need extra and tedious steps to be synthesized through traditional methods.

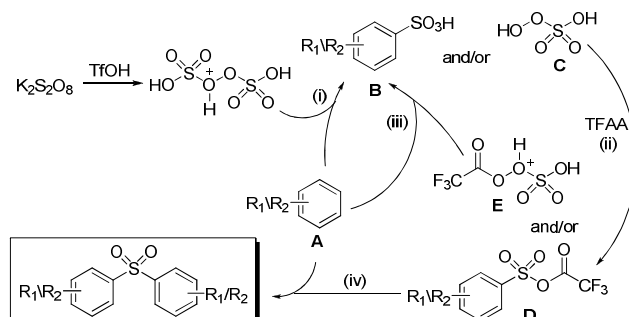
Table 3. Preparation of unsymmetrical diarylsulfones.

To further prove the practicality and applicability of this methodology, a gram-scale synthesis of some important chemicals were attempted. As shown in scheme 2, the intermediate **8** can be smoothly prepared in a large-scale (more than 10 grams) with an excellent yield. Remarkably, Dapsone (an antibacterial drug) can be readily accessed from simple bromobenzene in almost quantitative yield via a two-step chemical transformation. This route should represent the most efficient synthetic strategy towards Dapsone so far. Additionally, a few valuable diarylsulfone compounds (**48–52**) can be effectively constructed from compound **8** in excellent yields within a couple of steps. For instance, **51** and **52**, two useful building blocks in polymer science¹⁷, can be respectively obtained from intermediate **48** after a sequential operation including hydrolysis/aminolysis and hydrogenation. Meanwhile, these compounds can be further converted to heterocyclic sulfones.

Scheme 2. The gram-scale synthetic applications

Although details about the mechanism was not clear, on the basis of our observations, a plausible mechanism for this reaction

is depicted in scheme 3. Step (i) involves the aromatic sulfonation of arene **A** with TfOH and persulfate salts, which could afford the sulfonic acid **B** and potential sulfuroperoxoic acid **C**. In the second step (ii), intermediates **B** and **C** could be activated by TFAA to generate two kinds of mixed anhydride **D** and **E** respectively. The final step (iv) involves the second carbon-sulfur bond-forming reaction between **A** and **D** to furnish corresponding diarylsulfone products. Meanwhile, the trifluoroacetyl activated peroxosulfuric acid **E** can be captured by arene **A** to generate the second molecule of **B** in step (iii), which can undergo further steps (ii) and (iv) to provide diarylsulfones (more details see SI).

Scheme 3. The plausible mechanism

In summary, we have developed a new method to access diarylsulfones under transition-metal free conditions. Not only symmetric, also unsymmetric diarylsulfones can be readily prepared with simple arenes and persulfate salts. Persulfate salts, such as potassium persulfate, serve as the highly efficient sulfonation reagent and provide two sulfur atoms for C-S bond formation. Preliminary mechanism study reveals that TfOH and TFAA serve as the critical factors for C-S bond formation. The utility of this new reaction has been well exemplified in gram-scale synthetic applications. In comparison to traditional approaches, our new protocol has a few advantages including: 1) easily available simple arenes as the starting materials; 2) only one-step procedure; 3) without unpleasant odors during operations. Further studies into the scope of this reaction are in progress in our laboratory.

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